Assignment 1: Pulse Pressure Amplification Assessment

(EE6408: Non-invasive Sensing & Signal Processing for Human Health)

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Methods

Signal Processing Pipeline and beat detection:-

- The **raw signal** is first passed through a **moving average filter** to remove high-frequency noise and smooth the waveform.
- The filtered signal is then processed using a **minima detection filter** to identify local minima.
- A **sliding window** is used to find the minimum value in each window and compare it with the current sample point.
- If the current sample matches the **windowed minimum**, it is marked as a **local minimum**, and its **timestamp is stored**.
- These detected minima correspond to the **boundaries of cardiac cycles**, allowing **segmentation of individual beats.**
- The **time point's difference between consecutive minima** is calculated to determine the **beat interval**.
- Multiplying the beat interval by the **sampling period** gives the **beat duration**, which is then used to compute the **heart rate**.
- Using the detected minima, and limiting minimum beat duration to 0.3 s (200 bpm) the signal is **segmented into individual beats**.
- For each beat, the **mean, minimum, and maximum values** are calculated.
- Finally, these values are **averaged across all beats** to obtain the **final representative parameters for the dataset**.

Calibration Derivations:-

Full derivations are attached at the end of the document

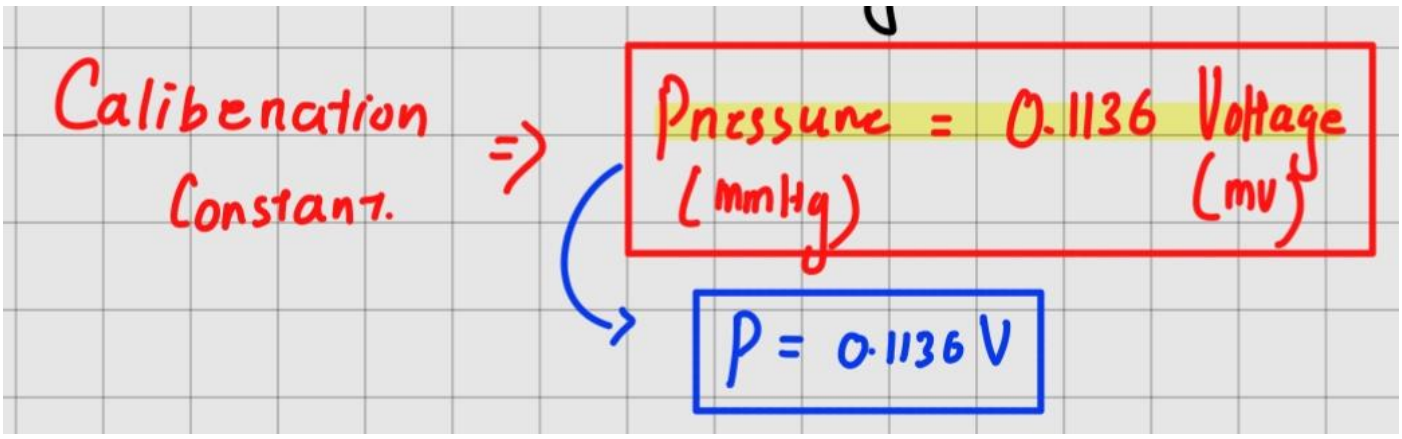


Figure 1-Calibration constant for Pressure values

$$P_{car}(t) = \frac{MAP - DBP}{x_{mean} - x_{min}} \left(x(t) - x_{min} \right) + DBP$$

Figure 2 Linear Calibration equation for Carotid pressure from diameter

Non linear Scaling

$$p(t) = P_{reff} + \frac{1}{\beta} \ln \left[\frac{D(t)}{D_{ref}} \right] \Bigg\} \rightarrow \text{Exponential}$$

$$P_{car}(t) \quad \frac{MAP_{brachial} - DBP_{brachial}}{MAP_{raw} - DBP_{raw}} \left(P_{raw}(t) - DBP_{raw} \right) + DBP_{brachial}$$

Figure 3 Nonlinear relation between input diameter and the pressure

Results

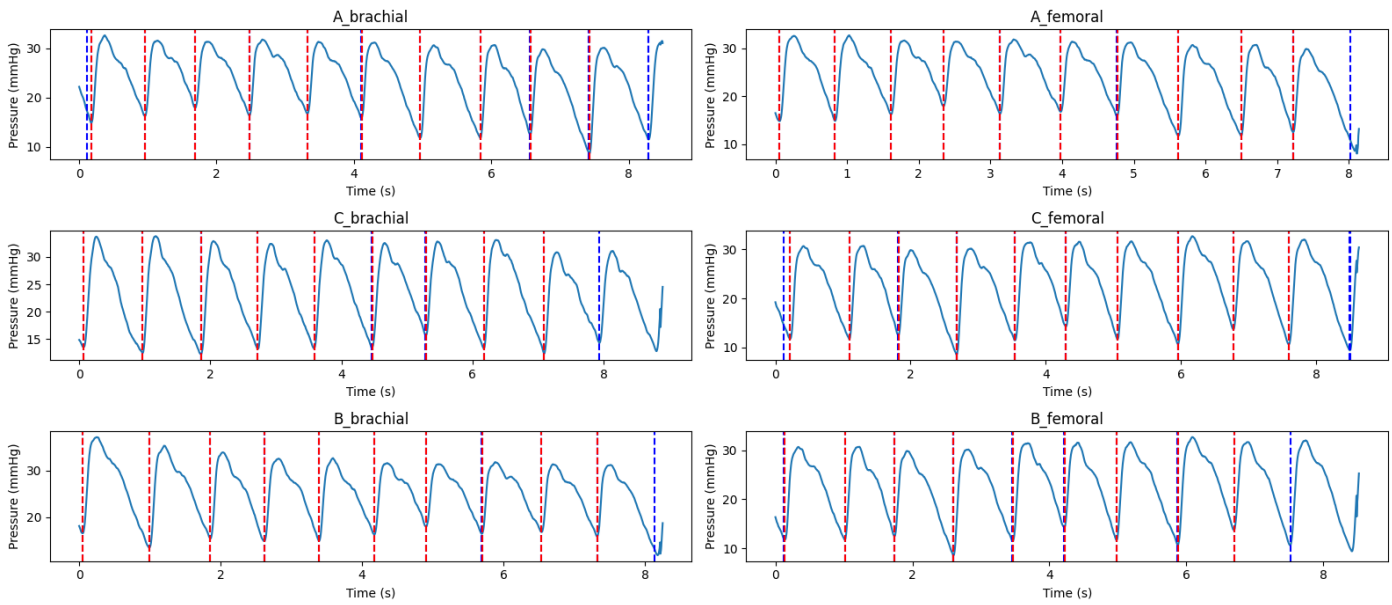


Figure 4- Plots showing waveforms of brachial and femoral blood pressure after calibration, filtering and with beats segmented using minima filter and beat duration limit.

SHEET NAME	SBP	DBP	MAP123	MAP	PP
A_brachial	24.58 mmHg	13.25 mmHg	17.02 mmHg	20.10 mmHg	11.33 mmHg
A_femoral	27.38 mmHg	14.54 mmHg	18.82 mmHg	22.26 mmHg	12.84 mmHg
C_brachial	27.18 mmHg	13.35 mmHg	17.96 mmHg	21.02 mmHg	13.83 mmHg
C_femoral	23.18 mmHg	10.85 mmHg	14.96 mmHg	18.31 mmHg	12.33 mmHg
B_brachial	28.74 mmHg	15.26 mmHg	19.75 mmHg	23.26 mmHg	13.48 mmHg
B_femoral	23.33 mmHg	11.04 mmHg	15.14 mmHg	18.54 mmHg	12.29 mmHg

Table 1- Different parameters like SBP, DBP, MAP for brachial and femoral blood pressure listed for 3 different data set

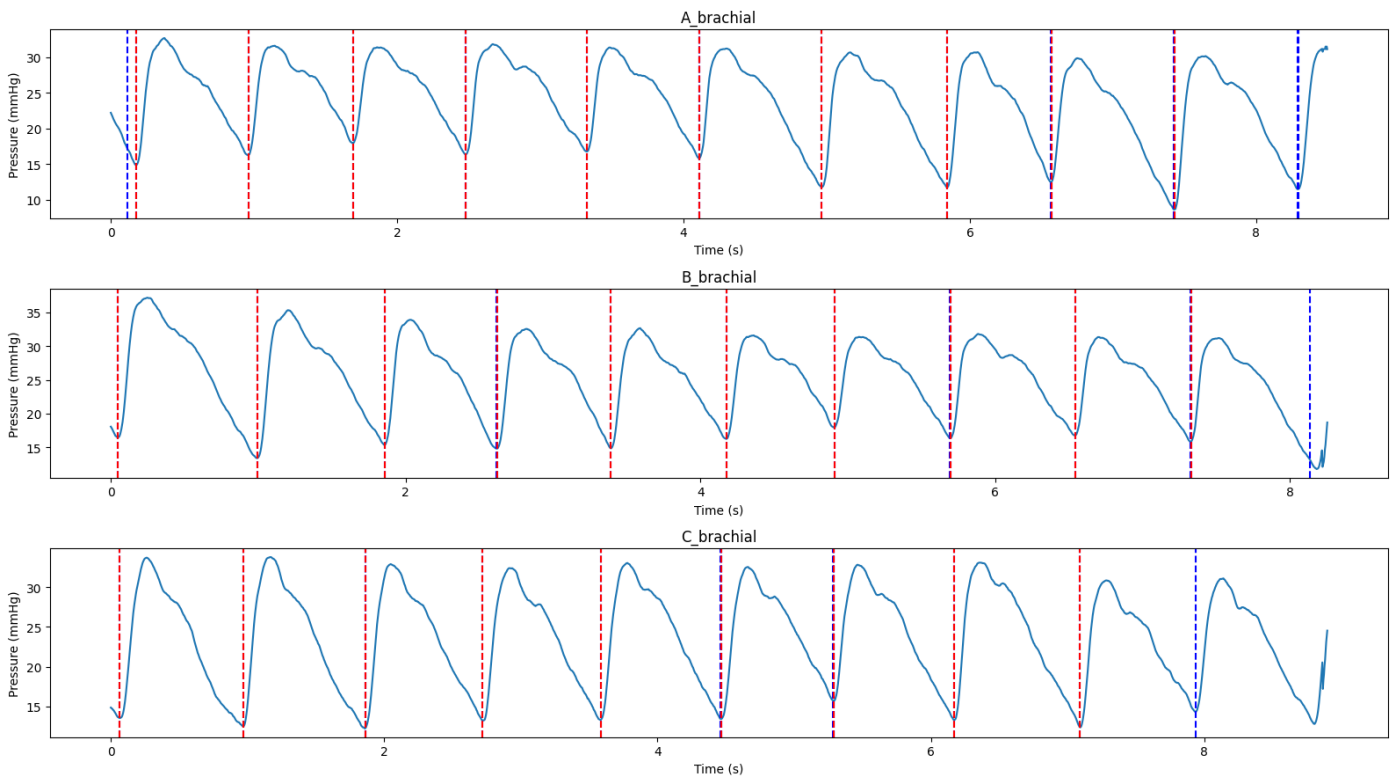


Figure 5 - Brachial Blood pressure waveform with beats detected and segmented.

Dataset	Avg Beat Interval (s)	Heart Rate (bpm)	SDNN (ms)	RMSSD (ms)	SD_SBP (mmHg)	SD_DBP (mmHg)	Observed Pattern
A_brachial	0.81	74.11	52.34	80.52	0.79	2.94	Higher HRV, moderate BP variability – resembles normotensive pattern
B_brachial	0.81	74.29	59.29	58.85	1.90	1.18	Highest SBP variability – resembles hypertensive-like pattern
C_brachial	0.87	68.74	29.68	45.28	0.81	0.48	Lowest HRV and BP variability – resembles hypotensive-like pattern

Table 2 - After beat detection parameters like average beat duration, Heart Rate, some statistical values and observed patterns

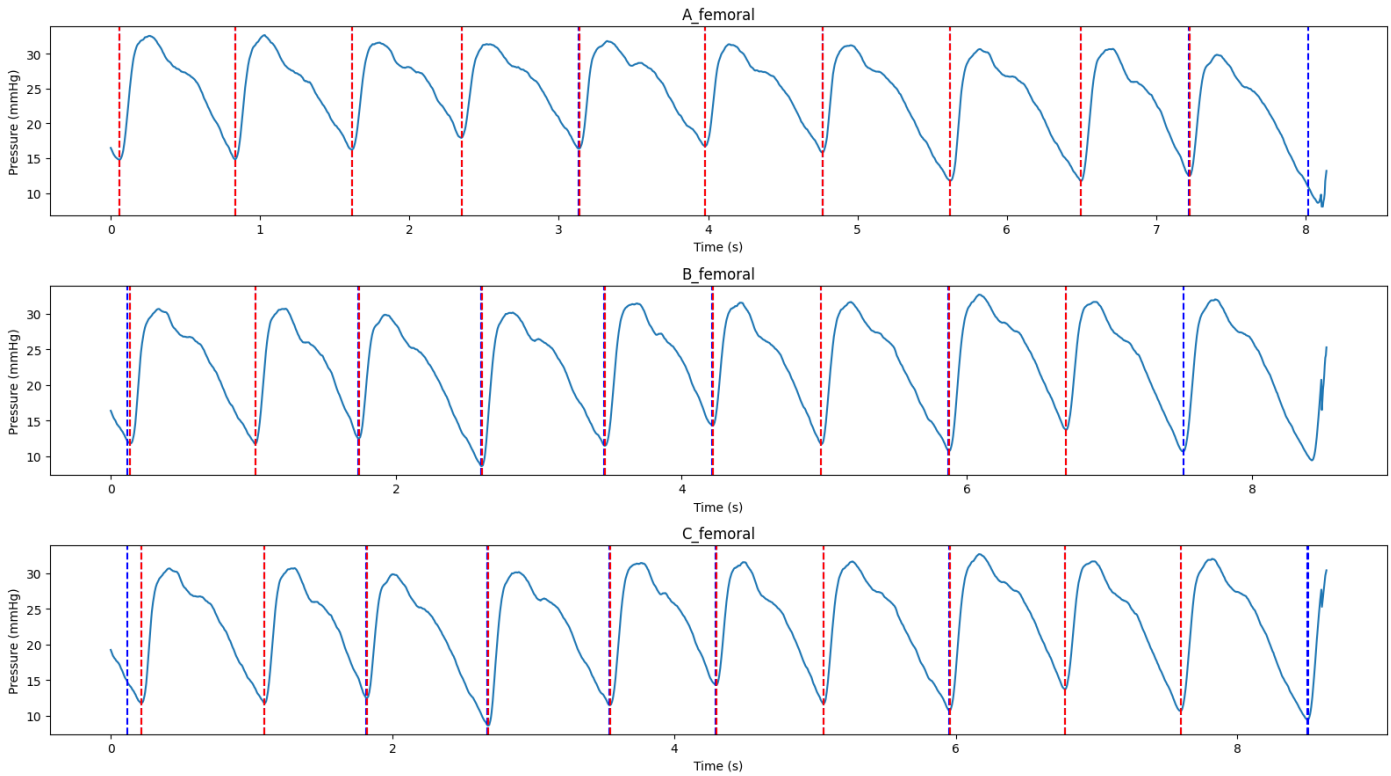


Figure 6 - Femoral Blood pressure waveform with beats detected and segmented.

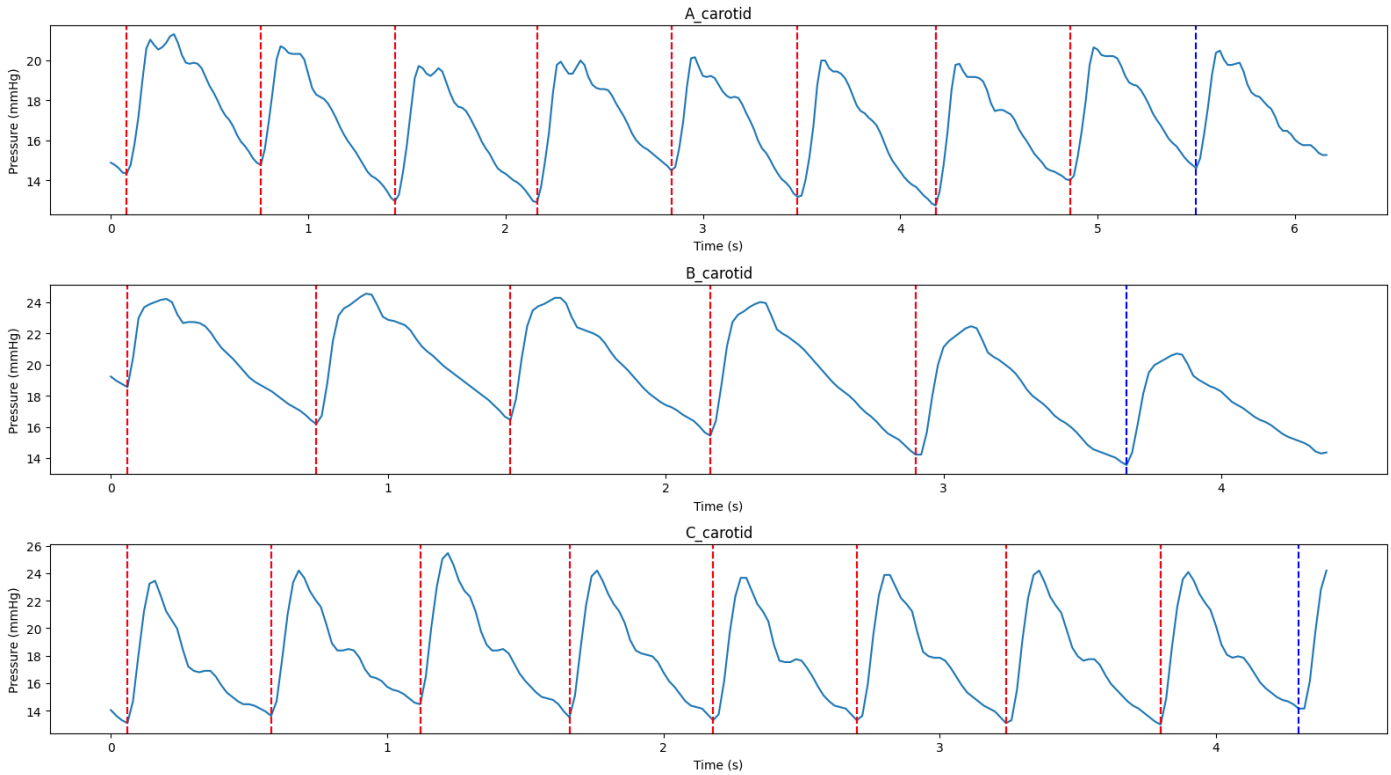


Figure 7 - Pressure waveform of carotid artery derived from linear relation with diameter with beats detected and segmented.

SHEET NAME	SBP	DBP	MAP123	MAP	PP
A_brachial	24.58 mmHg	13.25 mmHg	17.02 mmHg	20.10 mmHg	11.33 mmHg
A_femoral	27.38 mmHg	14.54 mmHg	18.82 mmHg	22.26 mmHg	12.84 mmHg
A_carotid	20.27 mmHg	13.09 mmHg	15.48 mmHg	16.95 mmHg	7.18 mmHg

SHEET NAME	SBP	DBP	MAP123	MAP	PP
B_brachial	28.74 mmHg	15.26 mmHg	19.75 mmHg	23.26 mmHg	13.48 mmHg
B_femoral	23.33 mmHg	11.04 mmHg	15.14 mmHg	18.54 mmHg	12.29 mmHg
B_carotid	22.72 mmHg	15.30 mmHg	17.77 mmHg	19.12 mmHg	7.42 mmHg

SHEET NAME	SBP	DBP	MAP123	MAP	PP
C_brachial	27.18 mmHg	13.35 mmHg	17.96 mmHg	21.02 mmHg	13.83 mmHg
C_femoral	23.18 mmHg	10.85 mmHg	14.96 mmHg	18.31 mmHg	12.33 mmHg
C_carotid	24.82 mmHg	13.09 mmHg	17.00 mmHg	17.90 mmHg	11.73 mmHg

Table 3 - SBP, DBP, MAP and PP for brachial, femoral and carotid blood pressure for different data set

SHEET NAME	PP_brachial / PP_carotid	PP_femoral / PP_carotid
A	1.58	1.79
B	1.82	1.66
C	1.18	1.05

Table 4 - Pulse Pressure Amplification for different data set t brachial and femoral with respect to carotid artery.

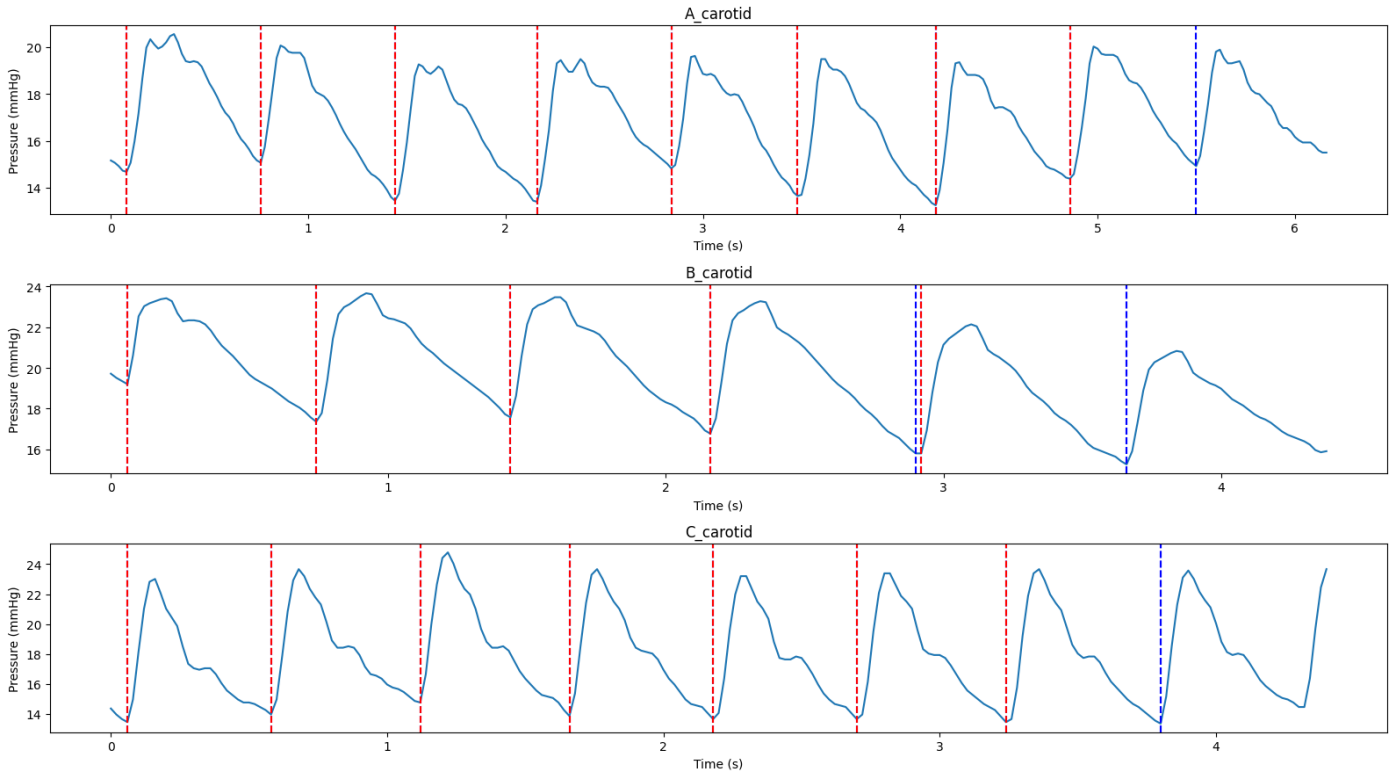


Figure 8 - Pressure waveform of carotid artery derived from non-linear exponential relation with diameter with beats detected and segmented.

Comparing Two Models

Linear Model

SHEET NAME	SBP	DBP	MAP123	MAP	PP
A_carotid	20.27 mmHg	13.09 mmHg	15.48 mmHg	16.95 mmHg	7.18 mmHg
B_carotid	22.72 mmHg	15.30 mmHg	17.77 mmHg	19.12 mmHg	7.41 mmHg
C_carotid	24.82 mmHg	13.09 mmHg	17.00 mmHg	17.90 mmHg	11.73 mmHg

Non-Linear Model

SHEET NAME	SBP	DBP	MAP123	MAP	PP
A_carotid	19.71 mmHg	13.57 mmHg	15.62 mmHg	16.92 mmHg	6.14 mmHg
B_carotid	21.23 mmHg	16.52 mmHg	18.09 mmHg	18.99 mmHg	4.71 mmHg
C_carotid	24.23 mmHg	13.45 mmHg	17.04 mmHg	17.95 mmHg	10.77 mmHg

Table 5 - Comparison between the 2 models used to model the carotid pressure waveform using diameter data set

Discussions

Q3. Pulse Pressure Amplification Across Arterial Sites

Pulse pressure (PP) differed across the brachial, carotid, and femoral arteries due to arterial wave propagation and reflection. In general, the carotid artery represents **central pressure**, while brachial and femoral arteries represent **peripheral pressures**. The results showed that PP was typically **lower in the carotid artery and higher in peripheral arteries**, demonstrating the phenomenon of **pulse pressure amplification**. This occurs because pressure waves travel through the arterial tree and are reflected at peripheral sites, which increases systolic pressure in distal arteries.

In the dataset corresponding to **normotensive conditions**, a relatively large PP amplification was observed between carotid and brachial/femoral sites. This is consistent with healthy and compliant arteries where wave reflections arrive later during systole. In contrast, the **hypertensive dataset showed reduced amplification**, which is associated with increased arterial stiffness. Stiffer arteries cause earlier wave reflections, increasing central systolic pressure and reducing the difference between central and peripheral pulse pressure. The **hypotensive dataset showed lower absolute pressures overall**, but the amplification pattern was still present, reflecting reduced vascular tone.

Overall, these results illustrate that **pulse pressure increases from central to peripheral arteries**, and the magnitude of amplification is influenced by arterial stiffness and vascular compliance. This behavior is consistent with known physiological patterns of arterial hemodynamics.

Q4. Linear vs Exponential Carotid Pressure Scaling

Two methods were used to estimate carotid pressure from the diameter waveform: **linear scaling** and a **single-exponential pressure–diameter model**. The linear scaling approach simply adjusts the waveform so that its diastolic and mean pressures match the brachial DBP and MAP. While this method is simple and computationally efficient, it assumes a linear relationship between arterial diameter and pressure, which may not accurately represent the mechanical behavior of arterial walls.

The exponential model, on the other hand, accounts for the **non-linear elastic properties of arteries**. Arterial walls become progressively stiffer at higher pressures, and the exponential model captures this behavior more realistically. In the results, the exponential method typically produced **slightly higher systolic pressures and larger pulse pressures** compared to the linear model. Additionally, the waveform shape generated by the exponential method showed augmentation that is more realistic **and reflected wave features**.

Across the datasets, the exponential method appeared **more physiologically plausible**, particularly in the hypertensive case where arterial stiffness is higher. The linear method tended to underestimate systolic pressure and reduce waveform curvature. Therefore, although both methods produced similar DBP and MAP due to calibration constraints, the **exponential model provided a better representation of arterial mechanics and central pressure dynamics**.

Q5. Brachial vs Central Blood Pressure and Hypertension Classification

Blood pressure classification is typically based on **brachial measurements**, with hypertension commonly defined as **SBP ≥ 130 mmHg or DBP ≥ 80 mmHg**. However, central pressures measured at the carotid artery can differ from brachial values due to pulse pressure amplification. In the analysis, brachial and carotid pressures did not always lead to identical classifications. In some cases, brachial systolic pressure was higher than central systolic pressure because peripheral amplification increases systolic pressure in the arm. This may lead to **overestimation of cardiovascular risk** when relying solely on brachial measurements. Conversely, in conditions with high arterial stiffness, central systolic pressure can become relatively high, potentially leading to **underestimation of central hemodynamic load** if only brachial pressure is considered. These findings highlight the importance of considering **central blood pressure** in cardiovascular risk assessment. Central pressure is more directly related to the load experienced by the heart and major organs. Therefore, methods that estimate central pressure, such as carotid waveform analysis, may provide **better insight into arterial stiffness, wave reflection, and true cardiovascular risk**.

EXTRA

Pulse Pressure Amplification

→ Pulse Pressure ⟶ difference between the Systole and Diastole pressure.

→ Amplification

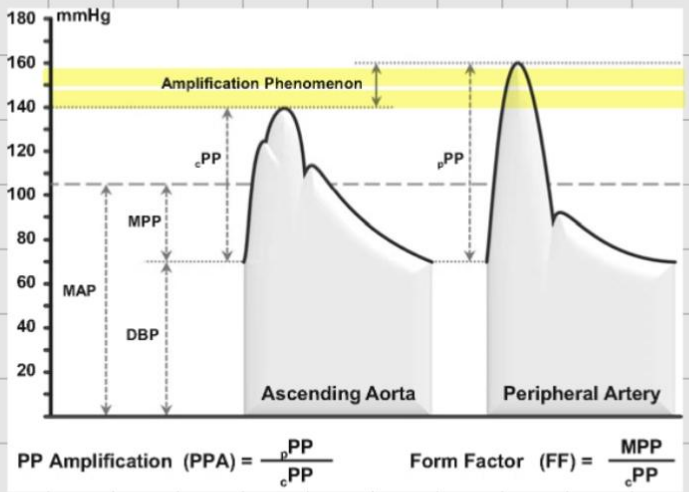
↳ https://ebrary.net/34843/health/pulse_pressure_amplification

as the pressure wave travels away from the central arteries (aorta) towards peripheri. (brachial, femoral) the pulse pressure ↑.

Why ↑

The pressure waves gets reflected due to (branching, impedance miss match, peripheral resistance) generating reflected waves.

↳ location & time



combines with the incoming waves. near (peripheral) faster and far (central) late. Rising the Systolic P at peripheral the diastolic P at central.

↳ causes the amplification of pulse pressure.

If stiffen arteries. faster reflection. Systolic wave ↑ instead of diastolic. This brings the (PPA) ↓.

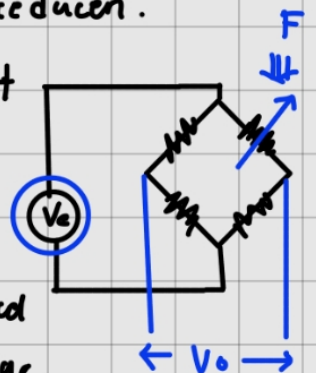
Sensor and Analog Frontend.

Sensor : Piezoresistive Bridge Pressure Transducer.

$V_o \propto V_e$ & $V_o \propto F$

Excitement Voltage

Applied Voltage.



Sensitivity = Δ Output / Δ input.

= 0.5 mv / V . KPA

V_{out} = 0.5 (mv/V KPA) . Δ input.

$$\boxed{V_{out} = 0.5 \times 3.3 \, (mv/KPA) \cdot \text{input}}$$

V_{out} → Sensor output

input → Applied pressure.

Before Amplification.

After amplification

$$\boxed{V_{out} = (40)(0.5)(3.3)/(7.5) \text{ input}}$$

V_{out} → Voltage

40 → Gain

0.5 → Sensitivity

3.3 → Exciting Voltage

7.5 → Kpa → mmHg.

So, $\boxed{V_{out} = 8.8 \, (\text{Pressure})}$

V_{out} → mv

Pressure → mmHg.

So Calibration Constant. ⇒ $\boxed{\text{Pressure (mmHg)} = 0.1136 \text{ Voltage (mv)}}$

$\boxed{P = 0.1136 V}$

$$t = \frac{\text{sample index}}{f_{sampling}} \Rightarrow t = i \times \frac{1}{250} \times \frac{4}{4}$$

$$\Rightarrow t = i\,(0.004) \Rightarrow \boxed{t = 4i \text{ ms}}$$

So the sensor is sampeled every 4 miliseconds.

Linear Scaling

$$\boxed{\overset{P}{y} = \underbrace{a}_{\text{Scaling}}\overset{\text{dia}}{x} + \underbrace{b}_{\text{bias.}}}$$

Range Mapping

given:— diastolic diameter.

= brachial DBP —— (I)

& mean level = brachial MAP. —— (II)

Applying (I)

$$a \cdot x_{min} + b = DBP_{(brachial)}.$$

Applying (II)

$$a \cdot x_{mean} + b = MAP_{(brachial)}$$

Subtracting (I) from (II)

$$a(x_{mean} - x_{min}) = (MAP - DBP)$$

$$\boxed{b = DBP - a\,x_{min}} \Rightarrow \boxed{a = \frac{MAP - DBP}{x_{mean} - x_{min}}}$$

$$P_{cap}(t) = a\,x(t) + b$$

$$= a\left(x(t) - x_{min}\right) + DBP$$

$$\Rightarrow \boxed{P_{cap}(t) = \frac{MAP - DBP}{x_{mean} - x_{min}}\left(x(t) - x_{min}\right) + DBP}$$

Non linear Scaling

$$p(t) = p_{reff} + \frac{1}{\beta} \ln \left[\frac{D(t)}{D_{ref}} \right]$$ → Exponential.

→ linear transformation

$$P_{cat}(t) = a \cdot P_{raw}(t) + b.$$ → linear Equation

① → $P_{cat,DBP} = DBP_{brachial}$

② → $P_{cat,MAP} = MAP_{brachial}$

given.

$$a = \frac{MAP_{brachial} - DBP_{brachial}}{MAP_{raw} - DBP_{raw}}$$

$$b = DBP_{brachial} - a\, DBP_{raw}$$

Obtained from ① & ② same as linear scaling

$$P_{cat}(t) = a\, P_{raw}(t) + DBP_{brachial} - a\, DBP_{raw}$$

$$\Rightarrow \boxed{P_{cat}(t) \quad \frac{MAP_{brachial} - DBP_{brachial}}{MAP_{raw} - DBP_{raw}} \left(P_{raw}(t) - DBP_{raw} \right) + DBP_{brachial}}$$

→ Non linear Equation after calibration.

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GITHUB LINK

https://github.com/shreenandansonu/Non-invasive-Sensing---Signal-Processing-for-Human-Health-IIT-Madras